

Matthew Holzer

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EDUCATION

The College of New Jersey (TCNJ)

September 2022 – May 2026

- Bachelor of Science in Computer Science, Minor in Digital and Creative Technology
- GPA: 3.7/4.0 | Dean's List: 6 Semesters

SKILLS

Languages: C, C++, C#, Java, Python, Ruby, TypeScript, JavaScript, SQL, HTML, CSS

Toolset: Git, GitHub, PostgreSQL, Linux, Unity, Unreal Engine, Blender, Maya

Frameworks & Libraries: React, Spring Boot, Spring AI, Ruby on Rails, MediaPipe, PyTorch, TensorFlow, PyGame

EXPERIENCE

Machine Learning & Computer Vision Researcher - *The College of New Jersey*

August 2025 – May 2026

- Conducted research on real-time American Sign Language recognition using machine learning and computer vision techniques, including model training, evaluation, and dataset analysis
- Improved handshape classification accuracy by 20% through model development, training, and evaluation on ASL datasets
- Designed and implemented a comprehensive testing pipeline to evaluate model performance, prediction accuracy, and system reliability

Computer Science Peer Mentor - *The College of New Jersey*

August 2024 – May 2026

- Tutored 80+ students in Java programming through one-on-one mentoring and lab instruction, helping students strengthen object-oriented programming and problem-solving skills

Fraternity President - *Kappa Theta Pi*

May 2024 – May 2025

- Led a technology-focused professional fraternity of 52 members, managing an 18-member executive board
- Revamped recruitment and organizational processes, increasing new member intake by 50% and improving member engagement
- Collaborated with national leadership on policy, programming, and organizational alignment

Virtual Reality Researcher - *The College of New Jersey*

Jan 2024 – May 2025

- Developed a virtual reality simulation in Unity (C#) to study ethical decision-making in a virtual environment
- Designed and implemented 10+ interactive VR scenarios in Unity using C#, object-oriented programming, and event-driven systems to support behavioral research studies

PROJECTS

Leave The World Behind: Strategy Hub

- Developed a full-stack companion platform using React and Spring Boot featuring discussion forums, chapter guides, and character wikis for an original tactical RPG project
- Integrated Google's Gemini API through Spring AI to analyze player discussion threads and generate AI-powered tactical summaries
- Engineered a RESTful backend using Java Spring Boot and PostgreSQL to efficiently map and query relationships between forum threads and nested comment chains

Legal Rights and Case Management Platform

- Built a secure full-stack web application using Ruby on Rails and PostgreSQL to inform LGBTQ+ individuals of legal rights, enable case reporting, and match clients with lawyers specializing in LGBTQ+ legal issues
- Developed a searchable database containing legislation from all 50 U.S. states, enabling rapid filtering and retrieval of relevant legal information

Leave The World Behind – Unity Project

- Developed a tactical RPG in Unity (C#) featuring grid-based combat and a flexible map system capable of supporting diverse mission objectives
- Implemented A* pathfinding and adaptive enemy AI to enable intelligent navigation and decision-making for player and enemy units
- Built a modular unit and item database, enabling future content additions without modifying core game logic

Nose-Tracking Drawing Application

- Developed a Python and PyGame drawing application controlled through real-time nose tracking using Google MediaPipe facial landmark detection